Neutrinos: A New Window into the Universe

GS III

- Traditional telescopes observe **electromagnetic radiation**, such as visible light and radio waves. However, some extreme cosmic environments are difficult to study using light alone because dense matter can absorb or scatter radiation before it reaches us.

- Neutrinos offer another way to investigate these environments. Because they have no electric charge and **interact very weakly** with matter, they can escape dense regions and travel enormous distances while preserving information about their origins.

- The **IceCube Observatory** detects these particles using Antarctic ice. When a neutrino occasionally interacts with matter, it can produce charged particles that generate faint flashes of light. Thousands of embedded sensors help scientists reconstruct its direction and energy.

- By combining neutrino observations with electromagnetic radiation, gravitational waves and cosmic rays, scientists practise **multi-messenger astronomy**. These different signals provide complementary evidence about violent cosmic events, helping researchers understand phenomena that conventional telescopes cannot fully explain.

Neutrino astronomy expands our ability to investigate the universe by detecting particles that carry information from extreme environments, complementing traditional observations based on light.